

CIS270A Project 9: Service Program

Now that you have practiced making modules, recall that they could be re-used again and again. This means they can be bound to other modules, become part of a service program, etc. **ILE!** (**ILE!** One should clap hands in air and shout “**ILE**” similar to the way one would shout an interjection to stir up excitement, such as “**OLE**”! This will set the tone for this project.)

In Project 8, you statically bound the modules together, with an entry point. This was done as a “bind-by-copy” process, which means that in the final product, your PRJ8MAIN *PGM object, all this code was combined/copied into the final product – a “statically bound” product.

In a service program, the process is to “bind-by-reference” which means that the code is not actually combined/copied, thus reducing redundancy. The service program can be used by any application, which will dynamically “grab” the subprocedures as needed at the time of the call. There is no designated entry point, as any module that is bound to the service program can become the entry point.

Next, you will create a service program called **USERSRVPGM** from your previously created modules **MODPASS** and **MODID**, and have an application utilize this service program. (Again, at this point, shout “**ILE!**”)

1. You should watch the “Service Program Demonstration Video” before starting.
2. Create the source physical file that will hold your binder language source member:
CRTSRCPF QSRVSRC
3. Create the binder language using either PDM or RDI, making sure the member type is **BND**.
4. Use **CRTSRVPGM** to create the service program, called **USERSRVPGM**. This service program will be comprised of your previously created **MODPASS** and **MODID** modules from Project 8. Because this is a long command, you will need to prompt and viewing the keywords will be helpful. This is demonstrated in the video.
5. The application that will use this service program is **USERAPP**, an interactive application that uses **USERSCREEN**. To keep the focus on the service program aspect:
 - a. **USERSCREEN** has already been coded and is in your QDDSSRC – study it and compile it in the usual manner
 - b. **USERAPP** has been mostly coded and is in your QRPGLSRC– study it, fill in the missing code and **CRTRPGMOD** to compile it, so that it becomes a module.
6. Next, you will bind the service program (remember this will “bind-by-reference”) to your **USERAPP** module to create a *PGM object, also named **USERAPP** that can be called:
 - a. **CRTPGM PGM(USERAPP) MODULE(USERAPP) BNDSRVPGM(USERSRVPGM)**
7. **CALL USERAPP** to test
8. Get out of your seat, and emphatically shout **ILE!** to celebrate your success. Do your housekeeping and send an email when done.